Psychometric genetics: a replication of Schwabe (2016)

Abstract

We partly replicated Schwabe (2016)'s findings that item-level models protect biometric inference from measurement artifacts and full information increases power with missing covariates: all 108 reconstructed precision contrasts favor full information, but the measurement-remedy and power claims are not established. Empirical effects could not be independently refitted, and repeated Bayesian recovery is incomplete. Eleven full-information bias point estimates reverse the source sign; none excludes zero after adjustment. For the illustrative heritability bias, the published positive effect is 0.03315402 versus our 0.03118039; its smaller magnitude does not imply equivalence. All 216 published variance-SE points fall outside conditional comparison bands; unknown source uncertainty and SE implementation prevent a definitive numerical replication verdict.

Scientific estimand / applicability	Published	Ours	Effect magnitude	Direction	Verdict
Item-subset invariance; proposition	not established	not established	not established	not established	partial
IRT bias remedy; joint claim	not established	not established	not established	not established	not established
Interaction power; design grid	not established	not established	not established	not established	not established
Sum-score artifacts; joint claim	not established	not established	not established	not established	not established
Mathematics AxE slope	2.0837	not established	not established	not established	blocked
Mathematics AxC slope	2.6159	not established	not established	not established	blocked
Two response dimensions; proposition	not established	not established	not established	not established	blocked
Nine-item subscale; proposition	not established	not established	not established	not established	blocked
Liberalism AxE slope	-2.81	not established	not established	not established	blocked
Liberalism AxC slope	0.54	not established	not established	not established	blocked
AxE sign reversal; joint claim	not established	not established	not established	not established	blocked
AxC sign reversal; joint claim	not established	not established	not established	not established	blocked
Logvariance identification; proposition	not established	not established	not established	not established	partial
Measurement-bias prevention; proposition	not established	not established	not established	not established	not established
Unique finite maximum; proposition	not established	not established	not established	not established	not established
Genetic power adequacy; proposition	not established	not established	not established	not established	not established
Shared-environment power; proposition	not established	not established	not established	not established	not established
20-item sufficiency; proposition	not established	not established	not established	not established	not established
SES shared-environment slope	-3.0777	not established	not established	not established	blocked
13 covariate nondetections; joint claim	not established	not established	not established	not established	blocked
Missing-data power; grid	not established	not established	not established	not established	not established
Variance-SE ordering; 108 contrasts	not established	not established	not established	not established	partial
Observed-data likelihood; proposition	not established	not established	not established	not established	partial
MAR A bias; 24 contrasts	not established	not established	not established	not established	partial
MAR C bias; 24 contrasts	not established	not established	not established	not established	partial
MAR heritability bias; 24 contrasts	not established	not established	not established	not established	partial
MCAR A SE: deletion minus full	0.00106356	0.00150235	not established	same direction	partial
MAR heritability: estimate minus truth	0.03315402	0.03118039	smaller	same direction	partial

Published values: Schwabe (2016), Tables 3.2, 4.3, 5.4 and Figures 6.3, 6.5. Slopes are log-variance change per source-scale unit; positive means increasing variance. Slope tolerances: 0.00005 (mathematics, SES), 0.005 (liberalism). The last two rows are illustrations, not pooled effects: MCAR, two covariates, 2% missing, A variance SE (squared phenotype units), tolerance 0.00023667095377570213; MAR, five covariates, 10% missing, heritability bias (proportion), tolerance 0.00027219389842495325. Positive means lower full-information SE or upward bias, respectively. Proposition/joint/grid rows have no scalar aggregate or magnitude tolerance. Descriptive point estimates; not equivalence. Source extraction bounds exclude unknown source Monte Carlo uncertainty. Unresolved SE comparability withholds magnitude despite the positive diagnostic contrast. Complete signed components and source locators are linked in Appendix A.

1 Introduction

We partly replicated Schwabe's (2016) conclusions about psychometric measurement and missing-covariate analysis, with the strongest numerical evidence supporting the Chapter 6 precision ordering rather than the full dissertation's empirical and power conclusions (Schwabe, 2016). All 108 deletion-minus-full variance-standard-error contrasts were positive with positive multiplicity-adjusted intervals. The illustrative heritability bias was positive in the source (0.03315402) and our reconstruction (0.03118039), and smaller by the geometric tolerance 0.00027219389842495325. Empirical refits, source-repetition Bayesian recovery and the reported full-information power advantage are not established. Eleven full-information bias point estimates reverse the source sign, but none excludes zero after multiplicity adjustment. A blocked comparison is not evidence of nonreplication.

The dissertation asks whether measurement properties can distort genetic and environmental inference, and whether joint modeling of incomplete covariates improves twin analyses. Its 2016 University of Twente version integrates item response theory (IRT) with additive-genetic, shared-environmental and unique-environmental components, denoted A, C and E. It studies latent genetic moderation, observed socioeconomic moderation and incomplete covariates. Chapter 7 qualifies the measurement argument: a common identified scale matters, and item modeling cannot repair poorly informative tails or systematic self-report bias (pp.107–108). We preserve those qualifications.

Our comparison covers seven chapters and Appendices A–G. The 996 linked source records include 26 principal claims and overlapping supporting quantities; they are not 996 independent studies. The first-page table retains all principal claims plus two preselected signed illustrations. Qualitative or compound claims have no invented scalar average. A separate scientific verdict can be partial even when a nonscalar magnitude is not established.

2 What we did ourselves, and what we took from others

Our contribution is an independent reconstruction and analytical audit, not a new empirical twin study. Schwabe supplied the model ideas, published designs, numerical summaries, figures and original program specifications. We computed conditional algebraic arguments, generated new fixed-seed data, implemented fits, evaluated convergence, and compared results with source quantities. Printed empirical estimates and digitized figure values remain attributed source evidence.

The Chapter 6 reconstruction comprises 48,000 datasets and 96,000 method fits, paired within datasets: 36 primary conditions and 12 separate sensitivity conditions, each with 1,000 datasets. All 96,000 fits passed the recorded numerical guards; none was discarded. Guard checks do not certify a global likelihood optimum. Response-generation checks for Chapters 2 and 5 comprise 2,650 datasets in 19 conditions and no posterior fits. Recomputing standardized reliability on those same datasets adds no independent datasets.

We used R 4.5.0 (R Core Team, 2025), OpenMx 2.21.13 for the Gaussian likelihood (Neale et al., 2016), and lavaan 0.6.19 for independent baseline comparisons (Rosseel, 2012). JAGS 4.3.2 with rjags 4-16 retains the non-Gaussian item-level likelihood (Plummer, 2024); a covariance-only substitute would change it. CODA 0.19.4.1 computes descriptive posterior summaries (Plummer et al., 2006), and posterior 1.6.1 supplies rank-normalized convergence and effective-sample-size diagnostics (Vehtari et al., 2021). Package versions and available execution identities are linked in the artifacts appendix. Ordinary mathematical checks were performed without a Lean certificate.

3 Does common measurement imply invariant biometric inference?

A common identified scale supports a restricted population-target result, but it does not guarantee identical finite-data evidence from different item subsets. Under calibrated conditionally independent binary items with positive logistic slopes, two distinct item observations identify the Gaussian latent mean and variance from population marginal and joint probabilities. Cross-twin joint probabilities then identify the covariance through a strictly positive Gaussian covariance derivative (Voigtlaender, 2020). At each supported moderator value,

$$A = 2(K_{\mathrm{MZ}} - K_{\mathrm{DZ}}), \qquad C = 2K_{\mathrm{DZ}} - K_{\mathrm{MZ}}, \qquad E = T - K_{\mathrm{MZ}}.$$

Both zygosities must occur at each of two distinct observed moderator levels to identify the stated log-linear coefficients. These are sufficient mathematical hypotheses, not verified properties of the unavailable applications.

Item subsets can give opposite finite likelihood orderings despite identifying the same population parameter. Consider positive ACE points $(1/4, 1/4, 1/2)$ and $(1/2, 1/8, 3/8)$, both with total variance one. DZ covariance is $3/8$ at both points, whereas MZ covariance increases from $1/2$ to $5/8$. With known Rasch items, one subset containing concordant successes favors the higher MZ covariance; another with success/failure patterns favors the lower covariance. Positive and negative derivatives of their pattern probabilities establish strict opposite orderings. This is not a claim about opposite global fitted optima or significance decisions, nor an assertion that the dissertation intended exact finite-sample equality.

The scope is narrower than universal IRT invariance. Chapter 2's latent-genetic moderation produces a non-Gaussian marginal model not covered by this conditional Gaussian argument. Ordinal and multidimensional observation kernels require their own identification work. Coherent affine changes of latent scale multiply variances by the squared scale factor; heritability is unchanged, but raw variances are not. Nonlinear transformations can change the model family. The complete argument and assumptions are in `handoff/scale-invariance-evidence.json`.

4 Empirical interactions and response scales

The original empirical interaction directions cannot be independently confirmed with the available inputs. Mathematics AxE was reported positive at 2.0837 with its 95% highest-posterior-density (HPD) interval excluding zero; AxC was 2.6159 with an interval containing zero (Table 3.2). Liberalism AxE was negative at -2.81 and AxC positive at 0.54, both with source HPD exclusions (Table 4.3). The socioeconomic shared-environment slope was -3.0777 (Table 5.4). These are source estimates, not our fitted effects. Original NTR/Cito linkage and calibration, Virginia responses, and relevant joint posterior inputs were not recovered by the documented public-source search.

The Chapter 4 dimensionality, nine-item subscale, and IRT-versus-sum-score sign reversals remain blocked empirical targets. The related published conservatism article and supplement provide model and response-category context (Schwabe et al., 2016); they do not supply the missing original responses. Source-only transformations of reported slopes or intervals cannot replace a joint posterior refit. Chapter 6's thirteen covariate nondetections at the reported 99.6% HPD criterion likewise cannot be reconstructed from the simulation grid. Nondetection is not a zero effect or equivalence claim.

Software behavior has a narrower evidential role. BayesTwin's pre-defense revision and version 1.0 were inspected separately (Schwabe, 2017). Its public example is simulated, not dissertation empirical data. Agreement of program semantics or replayed examples does not validate the empirical interaction findings. Each unavailable target, original value and locator is retained in the complete claim inventory.

5 Identification, variance functions and likelihood maxima

Log-variance moderation removes particular path-sign ambiguities and is conditionally identifiable, but finite-sample likelihood uniqueness is unresolved. With a shared binary moderator $m \in \{0, 1\}$, the squared path $(a + bm)^2$ has the same endpoint values after b is replaced by $-2a - b$. Even $a > 0$ leaves this reflection unless the endpoint is zero or additional positivity constraints are imposed. For three distinct moderator values, equality of squared affine paths implies equality up to global sign; a strictly positive intercept removes that global ambiguity. The binary and continuous statements therefore need different qualifications (Appendix C).

The exponential variance $V(m) = \exp(\beta_0 + \beta_1 m)$ is positive and has derivative $\beta_1 V(m)$. This proves monotonicity, including the constant zero-slope case, but not concavity of a likelihood. The Chapter 5 hierarchy yields $T = A + C + E$, $K_{\mathrm{MZ}} = A + C$ and $K_{\mathrm{DZ}} = A/2 + C$ under the stated independent innovations; the square-root scaling in Equation 5.14 is essential. Combining these moments with the calibrated observation assumptions establishes population identification and a unique population expected-log-likelihood maximizer.

A finite maximum need not exist for every possible dataset with a free mean. For all-zero binary responses, decreasing the fitted mean improves the likelihood toward an unattained supremum. This defeats a universal existence interpretation, but supplies no example of two finite maxima. The source's statement of "only one maximum" (p.72) is thus not established as an at-most-one finite-maximum result. A proper prior changes the objective; no posterior-unimodality or convergence conclusion follows. The full argument is in `handoff/chapter5-analytical-evidence.json`.

6 Missing covariates: precision, bias and power

Full information reproduced the qualitative precision ordering throughout the reconstructed Chapter 6 grid. Three missingness mechanisms, four covariate counts ($q = 2, 3, 4, 5$) and three primary rates (2%, 6%, 10%) supply 36 conditions; all 108 component-specific deletion-minus-full variance-SE contrasts were positive. Paired Student-t intervals with Bonferroni family size 108 excluded zero in every case (Dunn, 1961). For MCAR with $q = 2$ and 2% missingness, the A contrast was 0.00150235, MCSE 0.00002231, and adjusted interval [0.00142398,0.00158072], versus the source-derived 0.00106356.

The literal plotted variance-SE magnitudes were not reproduced within the conditional source-comparison bands. All 216 points lay outside bands combining our Monte Carlo uncertainty with geometric extraction bounds. Original Monte Carlo uncertainty and the source SE implementation are unavailable. Exploratory path-SE comparisons have only 20 points outside their corresponding bands, but path SE is a different estimand and cannot replace variance SE. Accordingly, the headline precision verdict is partial and its magnitude comparison unresolved. The figure's 3% label versus the 2% methods description is retained; 12 additional 3% conditions are a separate sensitivity family.

MAR deletion reproduced the reported bias directions across all 36 component-condition cells. A bias ranged from −0.072831 to −0.006218, C from −0.114958 to −0.007669, and heritability from 0.001416 to 0.031180. Adjusted intervals excluded zero in the expected direction in 10 of 12 A cells, all 12 C cells, and 9 of 12 heritability cells. Full-information biases were much smaller descriptively: A [−0.003192,0.001336], C [−0.002616,0.001017], and heritability [−0.001390,0.002654]. None excluded zero after adjustment, which does not establish unbiasedness or equivalence.

The heritability illustration preserves the contrast and its uncertainty. With MAR, five covariates and 10% missingness, deletion's mean estimate was 0.53118039 against truth 0.5, yielding bias 0.03118039; the source-derived estimate 0.53315402 yielded bias 0.03315402. Our MCSE was 0.00304897 and the ex-

ploratory 72-comparison adjusted interval [0.02080654,0.04155423]. The descriptive magnitude is smaller at the geometric bound 0.00027219389842495325, but this is not evidence of a statistically resolved source difference. Eleven full-information point signs reversed relative to the source; none had an adjusted interval excluding zero. Two conditional MAR source-estimate deviations occur at the ambiguous low-missingness level.

Saved-output checks address 21 additional descriptive Chapter 6 targets without adding fits or new hypothesis tests. All 81 adjacent-covariate-count deletion variance-SE point differences are positive. Absolute C deletion bias exceeds absolute A bias in all twelve MAR cells, with both raw and truth-relative comparisons retained. Under 10% MNAR missingness, full-information E means for two through five covariates are 0.208212, 0.207331, 0.206199 and 0.204763; the last is nearer 0.20 than 0.21. The recommendation-relevant subset has 54 positive pre-existing adjusted precision contrasts, but this does not establish an optimal "more than 3%, more than two covariates" policy. Recovery targets retain biases, MCSEs, RMSEs and between-method differences without inventing an equivalence margin. Five direct-power or unequal-structure targets remain pending.

Precision is not rejection-rate power. The source's general power assertion has no unique quantified alternative, test or rejection rule from which to reconstruct a power comparison. Neither lower SE nor a confidence interval containing zero establishes power or adequacy of twenty items. The Chapter 5 sample-size and item-count claims likewise require appropriate nonzero alternatives and detection fractions, not only posterior standard deviations under null moderation.

The observed-data likelihood and observation kernels were checked separately. The Gaussian covariance construction agrees with Equation 6.5 for $q = 2$–5. An exact conditional-variance example gives $2, 3/2, 12/11, 279/286$ as progressively more related observations are supplied, supporting how co-twin information can reduce uncertainty about a missing value. It does not guarantee smaller fitted parameter SE in every dataset. Appendix G's binary kernel uses probabilities 0.001 and 0.999 rather than a deterministic threshold; its continuous observation precision 100 adds variance 0.01 on the standardized scale. These exact semantics are documented in `handoff/chapter6-analytical-evidence.json`.

7 Response generation and Bayesian recovery

Generated response properties support some descriptive patterns but do not establish parameter recovery or power. Across 100 datasets each, the floor and ceiling designs had raw reliability means 0.908030 and 0.907533, versus the source descriptor 0.90. Mean skewness was positive 0.534467 and negative -0.543213, with the expected sign in all 100 datasets of each design. The source's skewness endpoints concern three estimators on one unknown-seed illustration; our replicate means are not that same estimand. The twelve Chapter 2 power-design generation cells had reliability means rounding to 0.92. The four Chapter 5 60-item designs rounded to 0.90, whereas the 20-item mean 0.738196 rounded to 0.74 against the reported 0.75. Neither favorable rounding nor the number of generated datasets supplies a posterior-fit result.

Changing the reliability formula did not explain the key descriptor gaps. Standardized alpha was 0.906973 for the floor design, 0.906451 for the ceiling design and 0.731098 for twenty items. Their paired standardized-minus-raw differences were -0.001057, -0.001082 and -0.007099 with MCSEs 0.0000570, 0.0000591 and 0.0001094. These compare formulas on the same 2,650 datasets; they are not two independent simulation samples. Source participant subset, aggregation and calibration conventions remain uncertain. All 19 conditions are retained below.

The fixed Chapter 2 floor posterior passes the stated operational diagnostic criteria, but supplies no repeated-dataset recovery result. Four adapted chains retained 28,000 draws each for the same seed-301 dataset with 500 twin pairs and 60 items. All fourteen monitored quantities passed the joint thresholds:

rank-normalized split/folded $\widehat{R} < 1.01$, bulk and tail effective sample sizes at least 400, and mean-MCSE/posterior-SD at most 0.05. The worst values were $\widehat{R} = 1.001673$, bulk ESS 685.762, tail ESS 1407.532, and relative mean MCSE 0.038211. Diagnostics follow Vehtari et al. (2021). Trace and rank inspection still showed serial dependence and rank differences in variance components; passing these criteria is not proof of convergence or an independent end-to-end posterior validation.

The interaction posterior mean was -0.123383, with pooled 95% HPD $[-0.684694, 0.430699]$. All twelve inspected whole- and half-chain HPDs included zero. This stable non-exclusion concerns one generated null-interaction dataset; it does not establish a zero effect, unbiasedness, calibrated detection or the source's 100-dataset detection fraction. Equal-tailed quantile MCSE is not HPD-endpoint MCSE. The shorter original and centered diagnostics failed their joint criteria. Every 12,000-draw centered prefix is identical to the corresponding prefix of the longer run, so those draws are not pooled as independent evidence. Ceiling, nonzero-interaction, larger-sample and Chapter 5 designs require separate validation. Exact summaries and dependencies are retained in `handoff/jags-centered-extended-recovery-scientific-evidence.json`.

8 Further conditional consequences of the models

Three additional analytical results clarify what the source models can support without establishing empirical performance. First, the Gaussian observed-moderator identification argument extends to disjoint MZ and DZ moderator supports if each support is not concentrated on three or fewer values and the same exact log-linear ACE functions apply across zygosities. A nonzero sum of at most four real exponentials has at most three distinct zeros, by an induction using Rolle's theorem. Equal conditional covariance functions at four support points must therefore agree everywhere. The ACE inversion then identifies the coefficients. This is a population result requiring calibrated items and available joint probabilities; it is not a guarantee of stable finite-sample extrapolation.

Second, Chapter 2's non-Gaussian latent model has an explicit fixed-parameter moment inverse. With $v_a > 0$, $v_c \geq 0$, $v_e > 0$ and interaction β, write

$$\Theta = \mu + A + C + \sqrt{v_e \exp(\beta A)}Z, \quad R = v_e \exp(\beta^2 v_a/2), \quad m_3 = 3v_a\beta R.$$

The twin covariances identify v_a, v_c as above, total variance gives $R = T - v_a - v_c$, and $\beta = m_3/(3v_a R)$, $\log v_e = \log R - \beta^2 v_a/2$. A nonzero interaction gives a nonzero third central moment and therefore a non-Gaussian marginal trait. This inverse identifies latent distributions on a fixed scale; no result shows that finite binary response probabilities identify all these moments. Fixed-parameter moment existence also cannot be exchanged with prior averaging: the source inverse-gamma priors have infinite means.

Third, exact Rasch score likelihood can retain the item model under known calibration. For responses y_j, score $S = \sum_j y_j$, and fixed difficulties b_j,

$$p(y \mid \theta, b) = \frac{\exp(S\theta - \sum_j b_j y_j)}{\prod_j (1 + \exp(\theta - b_j))}.$$

Summing patterns with the same score replaces $\exp(-\sum b_j y_j)$ by a coefficient depending only on the score and difficulties. Their ratio is independent of every latent-model parameter; integrating the same correlated latent law therefore preserves the normalized posterior. This does not justify a rescaled Gaussian sum-score comparator. Unknown calibration or discarded item masks break the argument, and no computational acceleration or recovery improvement has been measured. These three results are our conditional ordinary derivations, not additional source experiments; complete hypotheses are in `handoff/continuation-evidence-packet.json`.

9 Assumptions

9.1 Common calibration and observable joint probabilities

The population arguments assume known finite positive logistic slopes, a fixed anchored trait scale, conditional independence and sufficient observable within-person and cross-twin joint probabilities. Unknown calibration, informative item selection or absent joint observations require separate identification arguments. Source-scale invariance is not assumed across arbitrary transformations.

9.2 Gaussian shared-moderator reconstruction

The Chapter 5 and Chapter 6 covariance arguments assume the stated Gaussian innovations, common ACE functions across zygosity and known genetic correlations. The continuous-support extension additionally assumes exact log-linear functions across the relevant support. Neither the mixed-data application nor latent-genetic moderation is silently treated as this same Gaussian model.

9.3 Underreported generation choices

Chapter 5's reconstructed Bernoulli moderator probability is 0.5, consistent with the stated roughly 11% explained variance but not explicitly specified in the source. Chapter 2's fixed-versus-redrawn difficulty bank and original seeds are unresolved. The generation comparisons therefore test a declared reconstruction. No post-hoc choice of seeds, calibration or equivalence margin is used to force agreement.

9.4 Comparison precision and uncertainty

Rounding tolerance is half the source's last printed unit; for derived linear contrasts it is the sum of input bounds. Figure comparisons instead use declared coordinate-extraction bounds. A numerical match would round half up to the source value, close would differ by at most one last-place unit, and mismatch would exceed it, only for a comparable quantity. These labels differ from Monte Carlo uncertainty and reconstruction scope. Student-t Monte Carlo intervals use independent generated datasets with paired method differences; Bonferroni families retain the original dependence and target grouping. The full plan was specified during production, after baseline information was known, rather than wholly preregistered.

10 Ambiguities that were resolved

The analytical checks distinguish literal printed formulas from implemented likelihoods; some source inconsistencies remain unresolved rather than being silently repaired.

10.1 Environmental-variance ratio example (Chapter 2, p.21)

The variance-ratio identity is $\Delta = \exp(\beta_1\sqrt{V_A})$, so $\beta_1 = \log(\Delta)/\sqrt{V_A}$ for positive V_A. The neighboring 0.5 versus 0.2 baseline discrepancy does not change this identity; it does prevent treating both worked numbers as one common-baseline example.

10.2 Item information (Chapter 3, p.37)

For $p = \text{logit}^{-1}(a(\theta - b))$, information for θ is $a^2p(1-p)$. At $a = 2, \theta = b$, it is 1, whereas the printed formula gives 0.25. Unit slope resolves the narrow formula, but the source plot implementation is unavailable. Additivity requires conditional independence; inverse square-root information is not an arbitrary posterior SD.

10.3 Genetic shift convention (Chapter 3, pp.41–42)

The reported 1.63 and 1.85 ratios agree with one genetic standard deviation rather than variance. This identifies the numerical convention without independently recovering the empirical interaction posterior.

10.4 Category probabilities (Chapter 4, p.56; Appendix B)

The literal printed category probabilities sum to 1.25 at zero thresholds, while the Appendix B cumulative-exponential probabilities are normalized and equal 1/3 there. The code uses effective transitions indexed by the second and third thresholds. The normalized implemented target and the erroneous printed equations have different verdicts.

10.5 Heritability interval (Chapter 4, p.64)

The reported heritability mean 0.54 lies outside the printed HPD interval [0.34,0.51]. The inconsistency is retained; no corrected posterior interval is inferred without the original draws.

10.6 Sample size (Chapter 5, p.78; Table 5.1)

The table and results use 8,000 pairs while one design sentence says 80,000. The declared reconstruction uses the table design and preserves this discrepancy; the source design cannot be silently declared exact.

10.7 Separate-moderator appendix (Chapter 5, pp.74,84)

The separate-moderator program is Appendix E according to the contents and later text; the Appendix F reference is a locator discrepancy. Neither its source code nor this correction supplies extension-power evidence.

10.8 Low missingness (Chapter 6, pp.91–102)

The primary reconstruction uses 2% from the methods and a separate 3% sensitivity from plot labels. Both are retained, without selecting the better numerical agreement. The discrepancy affects the low-missingness comparison.

10.9 Application categories (Chapter 6, p.97; Table 6.1)

The enumerated and tabulated pedagogical categories are six, despite text saying five. The Reformed-liberated/liberal label discrepancy is retained in the empirical source mapping, not treated as a new measured group.

10.10 Covariate-count summaries (Chapter 7, p.104; Dutch summary, p.117)

The simulation includes five covariates as well as two, three and four. Summaries omitting the fifth condition do not justify dropping it from the full grid.

10.11 Shared-environment direction (Dutch summary, p.116)

The primary Chapter 4 table reports positive shared-environment moderation; the Dutch summary has the opposite wording. The table is the numerical source, and the translation is not a separate experiment.

10.12 Observation kernels (Chapter 6, p.97; Appendix G)

Appendix G uses the 0.001/0.999 binary kernel and continuous residual variance 0.01. These are exact probability-model differences from deterministic observation, not evidence that the empirical fit is wrong.

10.13 Identification and likelihood (Chapter 5, p.72; Appendix C)

Binary endpoint sign symmetry, continuous global sign symmetry, population identification and finite-likelihood uniqueness are separate statements. Removing path-sign ambiguity does not prove the last statement. The literal equation checks are supported by handoff/narrow-analytical-evidence.json; the source values and all thirteen ambiguity-to-claim mappings are retained in handoff/continuation-evidence-packet.json.

11 What could not be established

The dissertation's broad measurement-remedy, power and empirical conclusions are not settled by the completed narrow checks. Missing original data prevent empirical refits; incomplete repeated posterior experiments prevent source-count recovery and detection fractions; underreported procedures limit exact numerical comparison. Population injectivity does not establish a unique finite-sample optimum. The remaining core Bayesian source designs comprise 3,250 unique datasets and 3,450 dataset-method fits: 200 floor/ceiling datasets with two methods, 1,200 interaction-power/recovery datasets, 100 symmetric-comparator datasets, 1,000 sample-size datasets and 750 additional item-count datasets. These are target requirements, not completed fits. The 2,650 generation-only datasets omit the symmetric condition and the 100- and 250-item conditions; their count must not be interpreted as posterior completion. The following limitations retain the complete scientific scope of the evidence.

Limit 1. The extended fixed Chapter 2 floor case passes its operational diagnostic, but it does not reproduce the 100-dataset Chapter 2 or 250-dataset Chapter 5 summaries. Other source designs and repeated-dataset recovery remain unestablished.

Limit 2. Unavailable original empirical responses, calibration, linkage records, joint posterior draws and incompletely reported designs prevent the corresponding independent empirical and exact-design comparisons.

Limit 3. Figure extraction and arithmetic based on published parameters reuse source evidence. They are not independent data generation or empirical fits.

Limit 4. The ordinary mathematical arguments are conditional on explicit assumptions. They do not verify empirical calibration, observation mechanisms or support, and no formal proof certificate is supplied.

Limit 5. The 36 primary Chapter 6 conditions and 12 additional conditions with 3% missingness are distinct. The source describes 2% in prose and 3% in plots at the low level. Geometric extraction bounds are neither source Monte Carlo uncertainty nor equivalence margins.

Limit 6. Eleven full-information MAR bias point estimates reverse the source sign; none excludes zero after adjustment. Both conditional MAR source-estimate deviations occur at the ambiguous low-missingness level.

Limit 7. Shorter original and centered Bayesian diagnostics failed their criteria; the extended fixed case passes. This does not complete any 100- or 250-dataset source aggregate, detection fraction or empirical effect.

Limit 8. Population identification and common-scale arguments are assumption-bound; they do not establish finite-sample likelihood uniqueness. Ordinary analytical support is distinct from formal certification.

Limit 9. The 996 records include dependent components, parent claims and source-only arithmetic. They are not independent findings and do not define a replication success fraction.

Limit 10. The source reliability formula, aggregation and calibration are incompletely specified. Rounding agreement of reconstructed means is not an exact-source verdict; standardization does not explain the floor, ceiling or twenty-item gaps.

Limit 11. Qualitative constancy, slightness, closeness and adequacy have no justified equivalence margins. Descriptive Chapter 6 trends cannot establish those qualifications or a power advantage.

Limit 12. The conditional proofs do not establish finite-sample maximum existence or uniqueness, nor observed-response identification of the non-Gaussian latent moments.

Limit 13. All twelve whole- and half-chain interaction HPDs include zero for the fixed null-interaction dataset. Non-exclusion does not establish the null, unbiasedness or detection calibration. Quantile MCSE is not HPD-endpoint MCSE, and identical shorter prefixes are not pooled.

12 Where the reconstruction differs

The reconstruction has limited rather than exact scope wherever seeds, calibration, source aggregation or SE implementation are unknown. Its Gaussian missing-data design is fully fitted, but model guards are not a global optimum certificate. Generation-only comparisons and one-dataset posterior diagnostics do not stand in for the source's repeated recovery experiments. The full copyrighted dissertation and unlicensed original programs are locally inspected sources, not redistributed data. Conditional mathematical extensions add explicit assumptions and do not silently strengthen the dissertation's claims.

A Principal claims and complete linked inventory

Every source record, condition, original value, attempted calculation, uncertainty statement and remaining target is retained in handoff/continuation-evidence-packet.json and coverage-v4.csv. The 187 signed component comparisons are stored with their parent effects, including all 108 precision contrasts, 72 MAR bias contrasts and the retained interaction/power components. The 28 first-page rows do not replace them. This inventory is a set of dependent targets, not a replication success denominator. Exact-target accounting is 255 blocked, 391 pending, 29 attempted incomplete, 11 completed, 310 completed under declared reconstruction. In particular, 355 source-count aggregate targets remain pending. A completed diagnostic attempt is not a completed exact source target.

A.1 Under a common appropriately identified IRT measurement scale, biometric conclusions are asserted to be independent of the particular item subset.

C1.scale-invariance: partly replicated. Source: Chapter 1 p 5; Chapter 4 p 51; Chapter 7 pp 107–108. Conditional common-scale population identification is supported for the calibrated Gaussian model. Item subsets can produce opposite finite likelihood orderings; non-Gaussian latent-genetic and general ordinal models are outside that argument.

A.2 Incorporating an IRT measurement model remedies the scale-induced interaction bias and recovers generating parameters in the studied designs.

H2.IRT-bias-remedy: not established. Source: Chapter 2 abstract p 11; aim p 24; discussion p 25.. The complete multiscenario recovery claim is not established. One-data posterior checks and generation properties do not supply the source's repeated parameter-recovery summaries.

A.3 The proposed method has useful power to detect genotype–environment interaction under the investigated sample-size and effect-size conditions.

H2.power-grid: not established. Source: Chapter 2 abstract p 11; results p 22; discussion p 25.. The twelve source detection cells require 100 datasets per condition. Short pilots and one-data convergence checks do not yield those detection fractions; the curvilinear extension is incompletely specified.

A.4 Sum-score analysis can produce spurious genotype–environment interaction from measurement-scale properties.

H2.sumscore-artifacts: not established. Source: Chapter 2 abstract p 11; results pp 21–22; discussion pp 24–25.. Historical floor, ceiling and symmetric sum-score diagnostic slope means were 1.0951643, -0.8566462 and 0.1003354. These are one-dataset patterns, not the complete repeated source artifact comparison.

A.5 Unique-environmental variance increases with genetic mathematics ability; the AxE95% HPD excludes zero.

C3-4.3.2.beta1: blocked. Source: Chapter 3 p 42, Table 3.2. The source's mathematics AxE slope is 2.0837 with a 95% HPD exclusion of zero. Missing original responses, calibration and linkage prevent an independent empirical estimate.

A.6 The AxC point estimate is positive but its 95% HPD includes zero; shared-environment moderation is not detected.

C3-4.3.2.gamma1: blocked. Source: Chapter 3 p 42, Table 3.2. The source's mathematics AxC slope is 2.6159 with a 95% HPD interval containing zero. The original empirical inputs are unavailable; nondetection does not establish zero moderation.

A.7 The 28-item scale measures two distinct dimensions: liberalism–conservatism and use of the question-mark response.

C4-2.two-dimensions: blocked. Source: Chapter 4 abstract p 47, pp 58–61,Figs 4.1–4.4. The source interprets its 28-item scale as liberalism-conservatism and question-mark responding. The response data needed to re-estimate dimensionality are unavailable.

A.8 Nine selected items constitute the proposed ordinal social-liberalism subscale.

C4-3.subscale: blocked. Source: Chapter 4 pp 62,66. The source selects nine items for an ordinal social-liberalism scale. This methodological and empirical selection cannot be refitted from the published summaries alone.

A.9 Unique-environmental variance decreases with genetic liberalism; the negative AxE95% HPD excludes zero.

C4-5.4.3.beta1: blocked. Source: Chapter 4 p 63,Table 4.3. The source's liberalism AxE slope is -2.81 with an HPD exclusion of zero. No independent empirical posterior is available.

A.10 Shared-environmental variance increases with genetic liberalism; the positive AxC95% HPD excludes zero.

C4-5.4.3.gamma1: blocked. Source: Chapter 4 p 63,Table 4.3. The source's liberalism AxC slope is 0.54 with an HPD exclusion of zero. The positive primary-table direction is retained despite the contrary Dutch-summary wording; original data are unavailable.

A.11 The interaction point-estimate direction changes between IRT and sum-score analyses: beta 1

C4-5.reversal.beta1: blocked. Source: Chapter 4 pp 63–64,66–67,Tables 4.3–4.4. The source reports different AxE point-estimate signs under IRT and sum-score analyses. The two slopes have different scales, so their sign relationship is retained without inventing an average or subtraction; independent refitting is blocked.

A.12 The interaction point-estimate direction changes between IRT and sum-score analyses: gamma 1

C4-5.reversal.gamma1: blocked. Source: Chapter 4 pp 63–64,66–67,Tables 4.3–4.4. The source reports different AxC point-estimate signs under IRT and sum-score analyses. Independent empirical refitting is blocked, and the cross-scale sign proposition has no scalar aggregate.

A.13 The alternative logvariance ACE×M model is asserted to be uniquely identified.

C5-1.logvariance-identification: partly replicated. Source: Chapter 5 pp 69,72–77,83–84; AppendixC pp 138–139. Population identification is established under the explicit calibrated Gaussian observation assumptions, with separate shared-support and richer continuous-support results. The assumptions are not verified for unavailable empirical data. General finite-sample likelihood uniqueness is not implied; lack of a formal certificate is a separate certification limit.

A.14 Simultaneous IRT and ACE×M modelling is asserted to prevent spurious moderation arising from heterogeneous measurement error.

C5-1.measurement-bias: not established. Source: Chapter 5 pp 69,72–77,83–84; AppendixC pp 138–139. The method-level prevention of measurement-induced moderation is not established across the source designs. Exact Rasch score-likelihood sufficiency is a conditional likelihood result and does not validate the rescaled Gaussian sum-score comparator or repeated recovery.

A.15 The source states the logvariance model has only one maximum in its likelihood function.

C5-1.one-likelihood-maximum: not established. Source: Chapter 5 pp 69,72–77,83–84; AppendixC pp 138–139. Finite-sample at-most-one-maximum behavior is unresolved. An all-zero response example with a free mean lacks a finite maximizing point; nonexistence does not demonstrate multiple maxima. Population identification does not settle sample likelihood shape.

A.16 Large samples are said to be necessary to detect beta 1 a moderation.

C5-2.power-inference.beta1a: not established. Source: Chapter 5 abstract p 69,pp 78,83. The genetic-moderation sample-size adequacy assertion is not established. Posterior precision under null moderation does not supply power for a specified nonzero alternative or a sufficient-power threshold.

A.17 Large samples are said to be necessary to detect beta 1 c moderation.

C5-2.power-inference.beta1c: not established. Source: Chapter 5 abstract p 69,pp 78,83. The shared-environment-moderation sample-size adequacy assertion is not established. A detection experiment with a defined alternative and decision threshold is distinct from the source's uncertainty summaries.

A.18 As few as 20 phenotypic items are asserted sufficient to find moderation effects.

C5-3.twenty-sufficient: not established. Source: Chapter 5 pp 80,83. Twenty-item sufficiency for detection is not established. The generation-only reliability comparison and incomplete posterior evidence do not supply a defined power threshold or recovery guarantee.

A.19 Shared-environmental variance is lower for high-SES families; the negative C×SES95% HPD excludes zero.

C5-5.5.4.beta1c: blocked. Source: Chapter 5 p 82,Table 5.4. The source reports a shared-environment SES slope of -3.0777 with an HPD exclusion of zero. Original educational-response, SES and linkage inputs are unavailable for an independent estimate.

A.20 None of the thirteen measured covariate coefficients was detected in the educational-achievement application using the reported 99.6% HPD criterion.

H6.covariate-nondetection: blocked. Source: Chapter 6 abstract p 86; results p 99 and Table 6.3 p 100; discussion p 102.. The source reports no detection among thirteen covariate coefficients at its 99.6% HPD criterion. The linked empirical data and fitted-model inputs are unavailable; Chapter 6 simulation results cannot reproduce those application findings.

A.21 The source asserts that full information is more powerful than deletion regardless of the investigated missingness mechanism, covariate count or amount of missingness.

H6.power-assertion: not established. Source: Chapter 6 abstract pp 85–86; pp 91,101–102; Chapter 7 p 104.. The general greater-power assertion is not established. Chapter 6 supplies no uniquely quantified rejection-rate comparison with a defined nonzero alternative, and precision cannot substitute for it.

A.22 Full information gives lower variance-component standard errors than deletion across the investigated missingness mechanisms and design grid.

H6.precision-grid: partly replicated. Source: Chapter 6 results pp 93–95; discussion p 101.. All 108 reconstructed variance-SE differences favor full information with positive adjusted intervals. All 216 literal source variance-SE points differ outside conditional comparison bands; source SE implementation and Monte Carlo uncertainty are unresolved. The qualitative ordering is supported, but exact magnitude replication and power are not established.

A.23 The full-information model uses all observed phenotype and covariate components when covariate data are incomplete.

C6-1.observed-likelihood: partly replicated. Source: Chapter 6 pp 88–91;AppendixF. The Gaussian covariance construction and use of all observed components are supported analytically and computationally. This does not establish empirical distributional validity or universal finite-sample improvements, and the mixed-data application has additional observation kernels.

A.24 MAR deletion underestimates A variance while full information remains close to the generating value.

C6-3.MAR-bias.A: partly replicated. Source: Chapter 6 pp 94–95,Fig 6.5. Deletion A bias is negative in all twelve MAR conditions, with ten adjusted intervals excluding zero in that direction. Full-information bias is descriptively small, but intervals containing zero do not establish unbiasedness or equivalence.

A.25 MAR deletion underestimates C variance while full information remains close to the generating value.

C6-3.MAR-bias.C: partly replicated. Source: Chapter 6 pp 94–95,Fig 6.5. Deletion C bias is negative in all twelve MAR conditions and all twelve adjusted intervals exclude zero. Full-information zero bias is not established by its nonsignificant intervals.

A.26 MAR deletion overestimates heritability while full information remains close to the generating value.

C6-3.MAR-bias.h2: partly replicated. Source: Chapter 6 pp 94–95,Fig 6.5. Deletion heritability bias is positive in all twelve MAR conditions, with nine adjusted intervals excluding zero. Full-information estimates are descriptively near truth, without an established equivalence margin. The signed illustration and its uncertainty appear on page 1 and in the results.

A.27 Coverage by source section

The section totals below may overlap because an assertion can be used in multiple source locations. Repeated conclusions, cited background and future proposals are not counted as new experiments.

Source section	Coverage and exclusions
chapter_1; PDF pages 10–18; printed pages 1–9	7 linked claims. Cited textbook and external empirical results are background. Repeated dissertation conclusions retain the originating chapter claim IDs rather than being counted as new experiments.
chapter_2; PDF pages 20–36; printed pages 11–27	223 linked claims. All source-linked targets retained.
chapter_3; PDF pages 38–55; printed pages 29–46	54 linked claims. All source-linked targets retained.
chapter_4; PDF pages 56–77; printed pages 47–68	98 linked claims. All source-linked targets retained.
chapter_5; PDF pages 78–93; printed pages 69–84	206 linked claims. All source-linked targets retained.
chapter_6; PDF pages 94–111; printed pages 85–102	408 linked claims. All source-linked targets retained.
chapter_7; PDF pages 112–123; printed pages 103–114	8 linked claims. Cited textbook and external empirical results are background. Repeated dissertation conclusions retain the originating chapter claim IDs rather than being counted as new experiments.
appendix_A; PDF pages 140–142; printed pages 131–133	3 linked claims. All source-linked targets retained.

Source section	Coverage and exclusions
appendix_B; PDF pages 143–146; printed pages 134–137	8 linked claims. All source-linked targets retained.
appendix_C; PDF pages 147–148; printed pages 138–139	4 linked claims. All source-linked targets retained.
appendix_D; PDF pages 149–151; printed pages 140–142	1 linked claims. All source-linked targets retained.
appendix_E; PDF pages 152–155; printed pages 143–146	2 linked claims. All source-linked targets retained.
appendix_F; PDF pages 156–158; printed pages 147–149	2 linked claims. All source-linked targets retained.
appendix_G; PDF pages 159–163; printed pages 150–154	3 linked claims. All source-linked targets retained.
frontmatter; PDF pages 1–9	0 linked claims. Degree/title/edition metadata and table of contents; no original scientific result.
dutch_summary; PDF pages 124–126	0 linked claims. Translated summaries duplicate mapped Chapter 2–6 claims; no new experiment. Wording discrepancies retained in ambiguities.
bibliography; PDF pages 128–136	0 linked claims. Cited prior works are background; their scientific results are not original dissertation replication targets.
acknowledgements; PDF pages 138–138	0 linked claims. Acknowledgements of collaborators/funding, not original scientific results.
backcover; PDF pages 164–164	0 linked claims. Print production marks, not a scientific result.

B Generation comparisons in all 19 conditions

Published alpha descriptors come from Chapters 2 and 5 of Schwabe (2016); raw and standardized estimates below are our dataset means. Unknown source aggregation withholds exact-source numerical verdicts. The paired formula comparison changes no underlying observations. Condition numbering links to the complete scientific packet.

Cell	Chapter	Pairs	Items	Source alpha	Raw alpha	Standardized
1	2	500	60	0.90	0.908030	0.906973
2	2	500	60	0.90	0.907533	0.906451
3	2	500	60	0.92	0.915498	0.914910
4	2	1000	60	0.92	0.915660	0.915071
5	2	2000	60	0.92	0.915653	0.915039
6	2	500	60	0.92	0.915750	0.915064
7	2	1000	60	0.92	0.916238	0.915626
8	2	2000	60	0.92	0.915894	0.915258
9	2	500	60	0.92	0.916459	0.915872
10	2	1000	60	0.92	0.916481	0.915926
11	2	2000	60	0.92	0.916197	0.915574
12	2	500	60	0.92	0.917088	0.916535
13	2	1000	60	0.92	0.916215	0.915631
14	2	2000	60	0.92	0.917012	0.916441
15	5	1000	60	0.90	0.897739	0.894292
16	5	2000	60	0.90	0.897882	0.894467
17	5	4000	60	0.90	0.897706	0.894287
18	5	8000	60	0.90	0.897792	0.894389
19	5	1000	20	0.75	0.738196	0.731098

C Availability ledger

The dated public-source search below describes what was obtained, not proof that unavailable inputs do not exist. The ledger is copied from `handoff/scientific-availability.json`.

Date and source	Result
2026-10-30 Institutional final-version record	Verified PhD, institution, advisers, date, DOI, ISBN and final-version designation.
2026-10-30 Author dissertation GitHub repository	Full 164-page thesis and five model programs obtained at `6c2c48901209fc76ba932bb094c61abba5ef160d`. No raw empirical datasets or Monte Carlo driver/seeds. Full text read. Original files kept local; rights reserved/no code license.
2026-10-30 All 12 public author GitHub repositories	All cloned; immutable hashes in `author-repositories.json`. Data-file and code audit found workshop data, BayesTwin simulated posterior example and unrelated projects; no matching NTR/Cito/Virginia study inputs or curvilinear simulation driver. This is a search limit, not proof of absence.

Date and source	Result
2026-10-30 OSF indexed search	Queries site: osf.io Schwabe twin and exact Inga Schwabe 2016 found later unrelated projects/preprints, no matching dissertation simulation or empirical input. No identified thesis OSF project.
2026-10-30 GitLab indexed search	Queries Schwabe and exact ingaschwabe returned unrelated authors/no matching repository. No claim of exhaustive private-site search.
2026-10-30 Zenodo indexed search	Queries Schwabe twin and exact Inga Schwabe returned unrelated records/later projects; no matching dissertation data/code record.
2026-10-30 Chapter 4 article and supplement	Publisher confirms CC BY4.0 article and links ten-page supplement. Supplement accessible through browser; contains category plots, fit plots and JAGS program, not raw responses. Original thesis code replayed locally. Supplement findings cannot be refitted without Virginia 30 K responses.
2026-10-30 BayesTwin repository history	Current co `de656afd72d45ecd13f54b72c9400cceface8d423f`, version 1.0 dated 2017, GPL¿=2. Fetched full history; last pre-defense revision `a8e5d6b5ee6a8c76bdd0234a00fbcc809712ac08` dated 2015-10-15 available for separate replay. Example `results.rda` explicitly simulated, not dissertation empirical data.
2026-10-30 Historical statistical engine specification	Thesis cites R/rjags 2013 and OpenMx 2011/JAGS2003 publications but no complete sessionInfo, engine patch versions, random seeds or driver. Exact historical environment not determined. Current pinned versions and original-model replay used with explicit scope limit.

D Scientific compliance ledger

The following ledger is copied from `handoff/continuation-scientific-compliance.json`. Departures qualify the scope of the scientific evidence.

Requirement	Status	Evidence and justification
Whole-dissertation coverage	followed	The source inventory covers 19 sections, 996 dependent source records, 26 headline claims and 187 effect components. Exact-target work comprises 255 blocked, 391 pending, 29 attempted-incomplete, 11 completed and 310 completed under the declared reconstruction (`coverage-v4.csv`). Inventory coverage does not establish whole-study replication.
Original-source and input audit	followed	The input audit examined all 12 public author repositories. Original empirical responses, linked calibration and some simulation specifications were unavailable; their associated targets remain unresolved (`handoff/scientific-availability.json`).

Requirement	Status	Evidence and justification
Preferred statistical engines	followed	R 4.5.0, lavaan 0.6.19 and OpenMx 2.21.13 were available. OpenMx implements the Chapter 6 Gaussian likelihood, with independent baseline engine checks recorded in `outputs/engine-agreement.csv`; package versions are recorded in `provenance/preflight.json`.
JAGS for non-Gaussian latent item models	departed	A covariance-only SEM would change the source likelihood. JAGS 4.3.2 with rjags 4-16 reconstructs the item-level models; posterior 1.6.1 computes diagnostics. Generator, baseline engine and model-semantics checks do not independently validate the full posterior. The item-level posterior and its diagnostic limits are specified in `handoff/jags-gate-scientific-evidence.json` and `handoff/jags-centered-scientific-evidence.json`. The extended one-dataset result and runtime versions, including coda 0.19.4.1, are recorded in `handoff/jags-centered-extended-recovery-scientific-evidence.json`.
Original repetition counts and adequate posterior convergence	departed	Chapter 6 completes 1000 datasets in 36 primary and 12 sensitivity conditions, totaling 96000 method fits. Response generation completes 2650 datasets across 19 Chapter 2/5 conditions; alpha sensitivity adds zero independent datasets and zero posterior fits. Shorter original and centered diagnostics failed their joint criteria. The completed four-chain centered diagnostic with 28000 retained draws per chain passes the operational criteria for all 14 monitored quantities on one fixed Chapter 2 N500/J60 floor dataset; its interaction HPD includes zero. This does not establish the source 100/250-dataset posterior recovery or power aggregates, other designs, or empirical refits. `production/chapter6-v1/terminal-analysis/summary.json`; `handoff/generator-summary-scientific-evidence.json`; `handoff/generator-alpha-sensitivity-scientific-evidence.json`; `handoff/jags-centered-extended-recovery-scientific-evidence.json`.
Signed estimands, uncertainty and multiplicity	followed	The analysis distinguishes variance and path SEs, deletion-minus-full contrasts, estimate-minus-truth bias and source deviations (`production/analysis-plan.json`). Geometric extraction uncertainty, our Monte Carlo error and unknown source Monte Carlo error remain distinct. Descriptive magnitude is not equivalence.

Requirement	Status	Evidence and justification
Analytical verification and formal proof	departed	Ordinary mathematical arguments and exact algebra checks support the explicitly conditional results; no Lean certificate is claimed. Population identification, fixed-parameter latent moment inversion and exact Rasch score-likelihood sufficiency do not establish finite-likelihood uniqueness, non-Gaussian observed-item identification, empirical recovery, power or convergence. The derivations and assumptions appear in `handoff/chapter5-analytical-evidence.json`, `handoff/chapter6-analytical-evidence.json`, `provenance/continuous-moderator-identification-check.json`, `provenance/non-gaussian-latent-moment-check.json` and `provenance/rasch-sufficiency-derivation.json`.
Source attribution and permitted redistribution	followed	Schwabe (2016) is identified by verified bibliographic metadata and claim-specific dissertation locators. The full dissertation and unlicensed original programs remain local; redistributed materials comprise original replication code, generated results and limited attributed source extracts (`handoff/scientific-availability.json`).
Reproducible versions and commands	followed	Immutable repository revision `2424e0f4d67929ba72255ef6f546ba2a72c25df0` contains completed scientific outputs, diagnostic draws, reviewed checks and original replication code. Saved-output validation and fixed-diagnostic rerun commands are specified in `handoff/current-reproduction.md`; `handoff/terminal-reproduction.md` supplies the Chapter 6 recipe. Full copyrighted dissertation text and unlicensed original programs remain separate inputs.

E Artifacts and reproduction

The links below identify the actual filenames on the repository's master branch; they are navigation links rather than immutable versions. The Reproduction revision identifies `2424e0f4d67929ba72255ef6f546ba2a72c25df0`, which contains the code, completed Chapter 6 reports, generation summaries, mathematical checks and fixed recovery draws. From `formalization/schwabe-psychometric-genetics` in a separate checkout, follow `handoff/current-reproduction.md`. Saved-output checks and model-fitting commands are separate. The Chapter 6 analysis-only recipe is `handoff/terminal-reproduction.md`. Rebuilding that analysis requires Python, NumPy and SciPy and writes fresh output directories without fitting models. The packet's hashes identify the exact scientific evidence used here; the manuscript is a later artifact.

Artifact	Scientific role
`paper/standalone.tex`	Editable manuscript source
`paper/whole_dissertation_replication__schwabe_psychometric_genetics.pdf`	Rendered manuscript

Artifact	Scientific role
handoff/continuation-evidence-packet.json	All 996 claims, all signed effects, limits and source hashes
coverage-v4.csv	Exact-target and attempted-work status by source record
production/chapter6-v1/terminal-analysis/signed-SE-contrasts.csv	Paired variance and exploratory path SE contrasts
production/chapter6-v1/terminal-analysis/signed-bias-contrasts.csv	Signed MAR biases and uncertainty
production/chapter6-v1/terminal-analysis/source-comparisons.csv	Conditional source comparisons
production/chapter6-v1/terminal-analysis/condition-summary.csv	All parameter means, uncertainty and recovery diagnostics
production/analysis-plan.json	Estimands, denominator policy and multiplicity families
simulation/analyze_production.py	Saved-report analysis
simulation/build_production_report.py	Render saved analysis tables
simulation/openmx_production.R	Gaussian missing-data model implementation
handoff/terminal-reproduction.md	Pinned complete saved-analysis recipe
handoff/current-reproduction.md	Complete saved-output and fixed-diagnostic reproduction commands
outputs/generator-summary-analysis.json	All generation conditions and reliability/skewness summaries
outputs/generator-alpha-sensitivity-analysis.json	Paired raw versus standardized alpha
handoff/jags-centered-extended-recovery-scientific-evidence.json	Four-chain fixed-case diagnostic and scientific limits
handoff/chapter5-analytical-evidence.json	Identification and likelihood qualifications
handoff/chapter6-analytical-evidence.json	Covariance construction and observation links
handoff/narrow-analytical-evidence.json	Printed-formula and implemented-kernel checks
handoff/scale-invariance-evidence.json	Conditional population statement and finite-data example
provenance/runtime-packages.json	Pinned statistical runtime
provenance/posterior-runtime-packages.json	Diagnostic package versions
paper/references.bib	Editable bibliographic source

The original thesis PDF and source programs are local-only inputs. Re-extracting their values requires separately obtaining those originals; rebuilding saved tables does not. Repeating the statistical fits additionally requires the pinned R/OpenMx or JAGS runtime, full seed and protocol records, and a

fresh output namespace preserving all failures and denominators. The reproduction revision contains the scientific code and saved outputs; the manuscript and its evidence handoff have separate recorded hashes.

This manuscript can be rendered from the paper directory with `pdflatex standalone.tex`, `bibtex standalone`, and two further `pdflatex` passes. Copy the resulting `standalone.pdf` to `whole_dissertation_replication__schwabe_psychometric_genetics.pdf`. The bibliography is editable; all scholarly references below are used in the text.

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